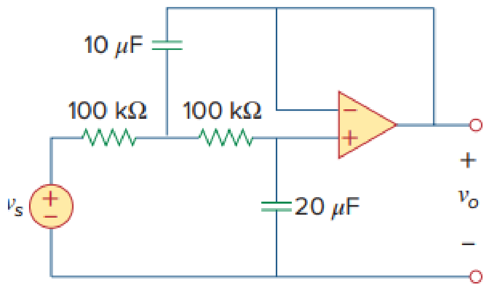
Problem

Obtain the differential equations for *vo*(*t*) in the op amp circuit of Fig. 8.111.

Figure 8.111:



Step-by-step solution

Step 1 of 3

Refer to the Figure 8.66 in the textbook,.

The circuit is a voltage follower therefore, the voltage across the capacitor $20\ \mu\text{F}$ is v_0.The circuit diagram with current directions at node 1 and 2 is shown in Figure 1.

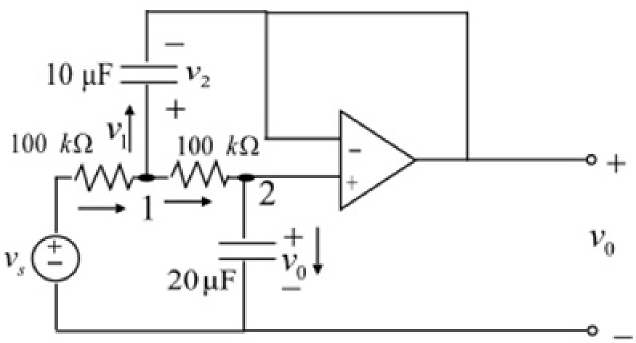


Figure 1

Step 2 of 3

Apply the Kirchhoff's current law at node 1.

$$\frac{v_s - v_1}{100\times10^3} = 10\times10^{-6}\frac{dv_2}{dt} + \frac{v_1 - v_0}{100\times10^3} \quad \text{...... (1)}$$

Apply the Kirchhoff's current law at node 2.

$$\frac{v_1 - v_0}{100\times10^3} = \left(20\times10^{-6}\right)\frac{dv_0}{dt} \quad \text{...... (2)}$$

Due to virtual ground, there will be no current flow in the operational amplifier.

Therefore,

$$v_2 = v_1 - v_0$$

Eliminate the values of v_2 and $\frac{v_1 - v_0}{100\times10^3}$ in equation (1).

$$\frac{v_s - v_1}{100\times10^3} = 10\times10^{-6}\frac{dv_1}{dt} - 10\times10^{-6}\frac{dv_0}{dt} + \left(20\times10^{-6}\right)\frac{dv_0}{dt} \quad \text{...... (3)}$$

Simplify the equation (2).

$$v_1 = v_0 + 2\frac{dv_0}{dt} \quad \text{...... (4)}$$

Step 3 of 3

Substitute the value of v_1 in equation (3).

$$\frac{v_s}{100\times10^3} = \frac{v_0}{100\times10^3} + \left(40\times10^{-6}\right)\frac{dv_0}{dt} + \left(20\times10^{-6}\right)\frac{d^2v_0}{dt^2}$$

$$\frac{d^2v_0}{dt^2} + \frac{dv_0}{dt} + 0.5v_0 = 0.5v_s$$

Therefore, the required differential equation for the output voltage v_0 is

$$\boxed{\frac{d^2v_0}{dt^2} + \frac{dv_0}{dt} + 0.5v_0 = 0.5v_s}.$$